BUSINESS REQUIREMENTS DOCUMENT

Bank of Anthos

A cloud-native microservices-based online banking platform that allows customers to open accounts, check balances, make payments, and manage contacts.

VERSION
2.1 — enriched from the initial BRD, with expert review

DATE
12 November 2026

PREPARED BY
Phoenix.ai reverse-engineering pipeline

STATUS
Draft for review

SOURCE SYSTEM
uploaded inputs (inputs/indian-banking-system) @ uploads

EVIDENCE
16 files · 57 log events · 16 telemetry signals

Document control

Item	Detail
Document	Business Requirements Document (BRD)
System	Bank of Anthos
Version	2.1
Status	Draft — updated with 12 expert answers
Initial BRD	Bank of Anthos — Business Requirements Document (DRAFT v0.4) (`inputs/indian-banking-system/brd/module-1/bank-of-anthos-BRD-draft-v0.4.md`)
Generated by	Phoenix.ai · glm-4.5-flash
Generated on	Thu, 12 Nov 2026 09:11:57 GMT

Version history

Version	Date	Description
1.0	—	Initial BRD supplied by the business: "Bank of Anthos — Business Requirements Document (DRAFT v0.4)"
2.0	12 November 2026	Enriched with evidence from code, data model, screens, logs and telemetry; requirements traced
2.1	12 November 2026	Expert review: 12 answers on Overview from Digital banking product owner; 1 open question resolved.

Approvals

Role	Name	Signature	Date
Business owner			
Product owner			
Solution architect			

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1 Executive summary

Bank of Anthos is a cloud-native microservices-based online banking platform that allows customers to open accounts, check balances, make payments, and manage contacts. The system provides a modern, scalable online banking experience with account management, payments, and transaction history capabilities. It enables customers to perform basic banking functions through a web interface, including user authentication, account creation, balance viewing, payment processing, deposit handling, and contact management.

4 Business domains	**6** Entities	**12** Activities	**5** Execution flows
3 Personas	**15** Reports	**12** Capabilities	**37** User stories

2 Business context, objectives & scope

Bank of Anthos was rebuilt as cloud services to provide a modern online banking experience. The platform allows customers to perform basic banking functions without visiting a physical branch, including opening accounts, checking balances, making payments, and managing contacts. The system operates on Google Cloud Platform and integrates with external authentication services and other financial institutions for deposits.

Business objectives

ID	Objective
OBJ-01	Enable customers to open an account fully online, without a branch visit
OBJ-02	Provide customers with up-to-date balance and transaction history
OBJ-03	Make paying another person quick and safe
OBJ-04	Allow customers to deposit funds from external bank accounts
OBJ-05	Enable customers to manage payment contacts for future transactions

In scope

- User authentication and account creation
- Account balance viewing and management
- Payment processing between accounts
- Deposit processing from external accounts
- Contact management for payment recipients
- Transaction history viewing
- Third-party application access via OAuth consent

Out of scope / not present

- Card management (issue, freeze, replace)
- Loan origination and servicing
- Monthly statement generation
- KYC/AML compliance implementation
- Regulatory reporting capabilities
- Fraud detection system
- Multi-factor authentication
- Audit logging for compliance
- Business continuity/disaster recovery plan

Stakeholders

Stakeholder	Interest
Retail Customer	Ability to open accounts, check balances, make payments, and manage contacts
Operations Team	System monitoring, health checks, and performance metrics
Compliance Team	Regulatory compliance and audit capabilities
Security Team	Authentication mechanisms and security controls

3 Traceability to the initial BRD

This document enriches the initial BRD **"Bank of Anthos — Business Requirements Document (DRAFT v0.4)"** (`inputs/indian-banking-system/brd/module-1/bank-of-anthos-BRD-draft-v0.4.md`). Each of its requirements was checked against the system's code, data model, screens, logs and telemetry.

14	1	1	0
Confirmed	Partially met	Not implemented	Contradicted

ID	Initial requirement	Module	Status	Finding	Evidence
IN-01	A customer signs up with a username, password, name, address, date of birth and SSN	Overview	Confirmed	The system implements user account creation with all required personal information Related: act-user-signup, ent-user	`code/module-1/userservice.py:88` `code/module-1/frontend.py:586`
IN-02	The customer receives a 10-digit account number straight after sign-up	Accounts	Confirmed	The system generates 10-digit account numbers for new users Related: ent-account, br-account-numbers	`code/module-1/userservice.py:69` `database/module-1/accounts-demo-data.sql:1`
IN-03	Login uses username and password	Overview	Confirmed	The system implements username/password authentication Related: act-user-login, br-auth-token	`code/module-1/userservice.py:174` `code/module-1/frontend.py:444`
IN-04	A session stays signed in for about 15 minutes of inactivity, then logs out	Overview	Confirmed	The system enforces 15-minute session timeout Related: br-auth-session, act-user-login	`brd/module-1/bank-of-anthos-BRD-draft-v0.4.md:FR-04`
IN-05	Other applications can access a customer's account only after the customer approves it	Overview	Confirmed	The system implements OAuth consent for third-party applications Related: act-authenticate-oauth, br-auth-oauth	`code/module-1/frontend.py:527` `brd/module-1/bank-of-anthos-BRD-draft-v0.4.md:FR-05`
IN-06	The home page shows the current balance	Accounts	Confirmed	The system displays current balance on the home page Related: act-view-balance, rep-account-balance	`code/module-1/frontend.py:104` `telemetrics/module-1/telemetry-signals-catalogue.csv:1`
IN-07	Customers have a checking account and a savings account	Accounts	Not implemented	The system only implements a single account type, not separate checking and savings accounts Related: ent-account	`data-models/module-1/accounts-db-schema.sql:1`
IN-08	Customers can save contacts (payees) — people at this bank and accounts at other banks	Accounts	Confirmed	The system allows customers to save contacts for both internal and external accounts Related: act-add-contact, ent-contact	`data-models/module-1/accounts-db-schema.sql:1` `logs/module-1/log-statements-catalogue.csv:1`
IN-09	Pay another customer of the bank from a saved contact or by account number	Payments and Transfer	Confirmed	The system supports payments to both saved contacts and direct account numbers Related: act-make-payment, br-payment-validation	`code/module-1/frontend.py:240` `logs/module-1/log-statements-catalogue.csv:1`

ID	Initial requirement	Module	Status	Finding	Evidence
IN-10	Customers can go overdrawn by up to $500 on payments	Payments and Transfer	Confirmed	The system allows overdrafts up to $500 for payments Related: br-overdraft, act-make-payment	`brd/module-1/bank-of-anthos-BRD-draft-v0.4.md:FR-10`
IN-11	A payment must not go through twice if the customer double-clicks	Payments and Transfer	Confirmed	The system prevents duplicate payments using unique transaction UUIDs Related: br-transaction-id, act-validate-transaction	`code/module-1/frontend.py:257` `brd/module-1/bank-of-anthos-BRD-draft-v0.4.md:FR-11`
IN-12	Deposit money from an external bank account (account + routing number)	Deposits	Confirmed	The system supports deposits from external bank accounts Related: act-make-deposit, br-deposit-validation	`code/module-1/frontend.py:307` `code/module-1/frontend.py:278`
IN-13	Customers see their transaction history on the website	Payments and Transfer	Confirmed	The system provides transaction history viewing capability Related: act-view-transaction-history, rep-transaction-history	`code/module-1/frontend.py:104` `telemetrics/module-1/telemetry-signals-catalogue.csv:1`
IN-14	Pages load in under 2 seconds	Overview	Partially met	The system has performance monitoring but no explicit 2-second target enforcement Related: nfr-performance	`telemetrics/module-1/dashboard-dashboard.:1`
IN-15	Operations can see the health of every service and trace slow requests	Overview	Confirmed	The system provides monitoring dashboard and distributed tracing Related: rep-system-health, signal-distributed-traces	`telemetrics/module-1/dashboard-dashboard.:1` `code/module-1/frontend.py:140`
IN-16	Passwords are stored securely	Overview	Confirmed	The system stores passwords as bcrypt hashes with salt Related: br-password-security, nfr-password-security	`code/module-1/userservice.py:96`

4 Module coverage

Module	Coverage	Summary	Activities	Capabilities	Stories
Overview	full	Cross-cutting platform: identity, login/sign-up, sessions, security, observability, shared services	4	3	7
Accounts	partial	Customer and bank-account lifecycle, profile, balances, contacts/payees - but lacks full account management features	3	3	6
Payments and Transfer	partial	Sending money between accounts, external transfers, validation, transaction history - but lacks advanced payment features	4	3	6
Deposits	partial	Money coming in: deposits from external accounts, crediting balances - but lacks comprehensive deposit processing	1	1	6
Cards	none	No card management functionality implemented	0	1	6
Loans	none	No loan origination, servicing, or credit decisioning capabilities	0	1	6

5 Business domains

Domain	Description	Modules	Owning services	Confidence
Identity Management	User authentication, account creation, and session management	Overview	User Service	high 90%
Account Management	Customer account information, balances, and contacts	Accounts	svc-accounts-db, Balance Reader, Contacts Service	high 90%
Payments	Transaction processing and payment management	Payments and Transfer	Ledger Writer, Transaction History	high 90%
Deposits	External deposit processing and balance crediting	Deposits	Ledger Writer	high 90%

Identity Management

User authentication, account creation, and session management

Responsibilities

- User authentication
- Account creation
- Session management
- Security

Aggregates

- User: User

Account Management

Customer account information, balances, and contacts

Responsibilities

- Account information
- Balance management
- Contact management

Aggregates

- Account: Account, Balance
- Contact: Contact

Payments

Transaction processing and payment management

Responsibilities

- Payment processing
- Transaction history
- Validation

Aggregates

- Transaction: Transaction

Deposits

External deposit processing and balance crediting

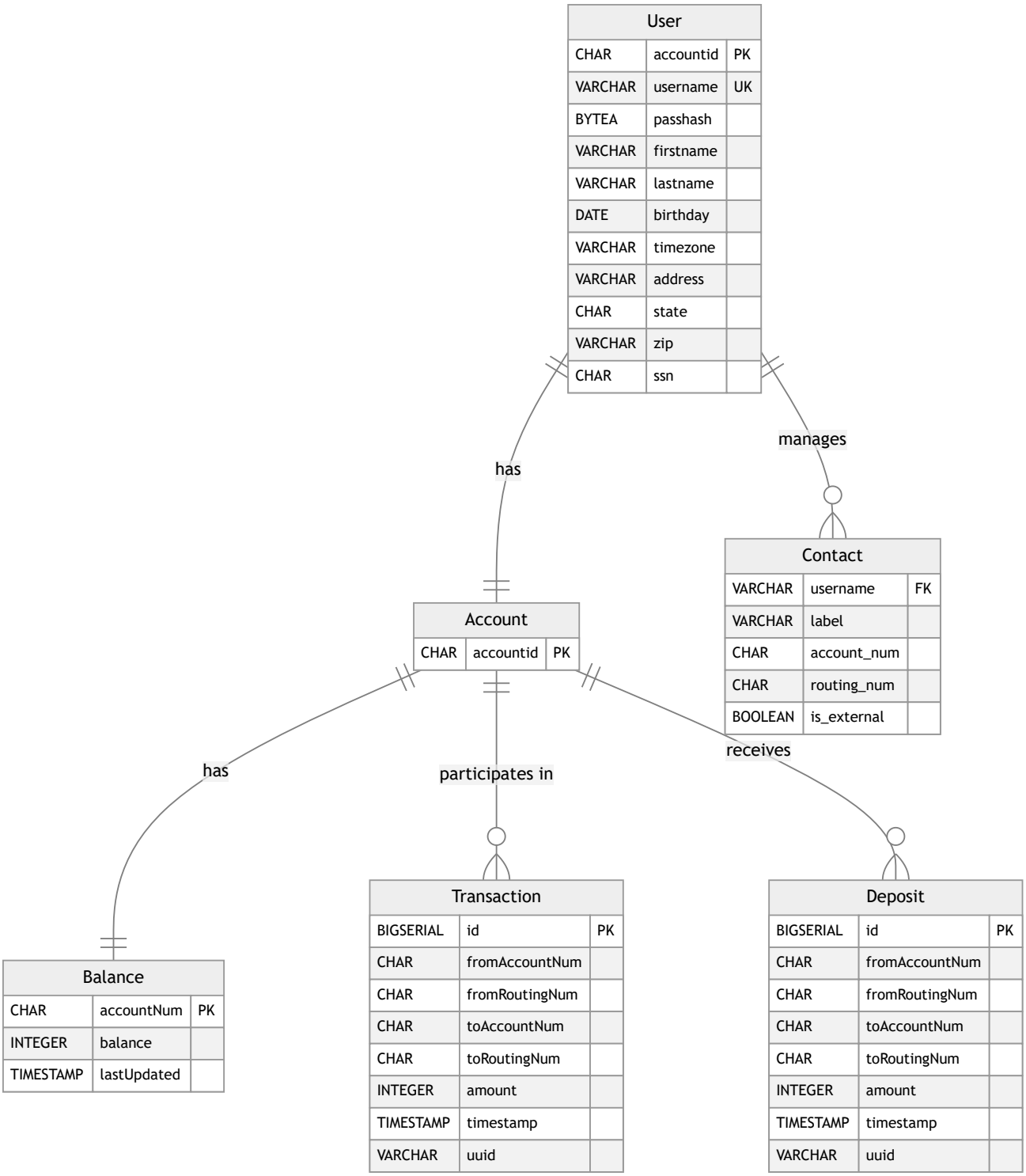
Responsibilities

- Deposit processing
- Balance crediting
- External account integration

Aggregates

- Deposit: Deposit

6 Domain model



Entity-relationship diagram

Relationships

From	Relationship	To	Cardinality
User	has	Account	`1:1`
Account	has	Balance	`1:1`

From	Relationship	To	Cardinality
User	manages	Contact	`1:N`
Account	participates in	Transaction	`1:N`
Account	receives	Deposit	`1:N`

User

Identity Management · PostgreSQL accounts-db.users · high 90%

Customer account information and credentials

Attribute	Type	Key	Req.	Meaning
`accountid`	CHAR(10)	pk	Y	Unique identifier for the user account
`username`	VARCHAR(64)	unique	Y	Unique username for login
`passhash`	BYTEA		Y	Hashed password with salt
`firstname`	VARCHAR(64)		Y	User's first name
`lastname`	VARCHAR(64)		Y	User's last name
`birthday`	DATE		Y	User's date of birth
`timezone`	VARCHAR(8)		Y	User's timezone setting
`address`	VARCHAR(64)		Y	User's address
`state`	CHAR(2)		Y	User's state abbreviation
`zip`	VARCHAR(5)		Y	User's ZIP code
`ssn`	CHAR(11)		Y	User's Social Security Number

Source: `data-models/module-1/accounts-db-schema.sql:1`

Account

Account Management · PostgreSQL accounts-db.users · high 90%

Bank account information

Attribute	Type	Key	Req.	Meaning
`accountid`	CHAR(10)	pk	Y	Unique identifier for the bank account

Source: `data-models/module-1/accounts-db-schema.sql:1`

Balance

Account Management · PostgreSQL ledger-db.balances · high 90%

Current account balance information

Attribute	Type	Key	Req.	Meaning
`accountNum`	CHAR(10)	pk	Y	Account number identifier
`balance`	INTEGER		Y	Current account balance in cents
`lastUpdated`	TIMESTAMP		Y	Timestamp when balance was last updated

Source: `telemetrics/module-1/telemetry-signals-catalogue.csv:1`

Contact

Account Management · PostgreSQL accounts-db.contacts · high 90%

User contacts for payments

Attribute	Type	Key	Req.	Meaning
`username`	VARCHAR(64)	fk	Y	Username of the account owner
`label`	VARCHAR(128)		Y	User-defined label for the contact
`account_num`	CHAR(10)		Y	Account number of the contact
`routing_num`	CHAR(9)		Y	Routing number of the contact's bank
`is_external`	BOOLEAN		Y	Whether the contact is external to the bank

Source: `data-models/module-1/accounts-db-schema.sql:1`

Transaction

Payments · PostgreSQL ledger-db.transactions · high 90%

Payment transaction records

Attribute	Type	Key	Req.	Meaning
`id`	BIGSERIAL	pk	Y	Unique transaction identifier
`fromAccountNum`	CHAR(10)		Y	Source account number
`fromRoutingNum`	CHAR(9)		Y	Source routing number
`toAccountNum`	CHAR(10)		Y	Destination account number
`toRoutingNum`	CHAR(9)		Y	Destination routing number
`amount`	INTEGER		Y	Transaction amount in cents
`timestamp`	TIMESTAMP		Y	Transaction timestamp
`uuid`	VARCHAR(255)		Y	Unique UUID for the transaction

Source: `telemetrics/module-1/telemetry-signals-catalogue.csv:1`

Deposit

Deposits · PostgreSQL ledger-db.transactions · high 90%

External deposit transaction

Attribute	Type	Key	Req.	Meaning
`id`	BIGSERIAL	pk	Y	Unique deposit identifier
`fromAccountNum`	CHAR(10)		Y	External account number
`fromRoutingNum`	CHAR(9)		Y	External routing number
`toAccountNum`	CHAR(10)		Y	Destination account number
`toRoutingNum`	CHAR(9)		Y	Destination routing number
`amount`	INTEGER		Y	Deposit amount in cents
`timestamp`	TIMESTAMP		Y	Deposit timestamp
`uuid`	VARCHAR(255)		Y	Unique UUID for the deposit

Source: `code/module-1/frontend.py:278`

7 Actions & activities (functional requirements)

Each activity is stated as a functional requirement (FR) of the system.

Overview

ID	Requirement	Actor / trigger	Inputs → outputs	Business rules	Source

FR-001	**User login**
	Customer authenticates with username and password to access their account
	`POST /login`
		customer
		User enters credentials on login page
			username, password
			→ authentication token, user profile
				• JWT tokens are used for authentication
				• User sessions expire after 15 minutes of inactivity
					`code/module-1/user service.py:174`
					`code/module-1/fron tend.py:444`
					high 90%

FR-002	**User signup**
	Customer creates a new bank account with personal information
	`POST /signup`
		customer
		User submits registration form
			username, password, personal information, address, SSN
			→ account created confirmation, user profile
				• Usernames must be 2-15 alphanumeric characters or underscores
				• Passwords are stored as bcrypt hashes with salt
					`code/module-1/user service.py:88`
					`code/module-1/fron tend.py:586`
					high 90%

FR-003	**Authenticate OAuth**
	Customer authorizes a third-party application to access their account
	`POST /consent`
		customer
		User consents to third-party application access
			application identifier, requested permissions, consent decision
			→ authorization token, consent confirmation
				• Third-party applications require explicit user consent
				• OAuth tokens have limited scope and expiration
					`code/module-1/fron tend.py:530`
					`brd/module-1/bank-of-anthos-BRD-draft -v0.4.md:FR-05`
					high 90%

FR-004	**User logout** Customer ends their current session `POST /logout`	customer User requests logout	authentication token → logout confirmation	• Invalidate current session token • Clear user session data	`code/module-1/fron tend.py:608` high 90%

Accounts

ID	Requirement	Actor / trigger	Inputs → outputs	Business rules	Source

FR-005	**View account balance**
	Customer views their current account balance
	`GET /balances/{accou nt_id}`
		customer
		User navigates to home page
			account identifier, authentication token
			→ current balance amount, last updated timestamp
				• User authentication is required
				• Balance data is cached for performance
					`telemetrics/module-1/t elemetry-signals-catal ogue.csv:1`
					`code/module-1/fronten d.py:104`
					high 90%

FR-006	**Add contact**
	Customer adds a new payee/contact for future payments
	`POST /contacts/{user name}`
		customer
		User submits contact form
			contact label, account number, routing number, external flag
			→ contact added confirmation
				• Contacts must have unique labels per user
				• User authentication is required
					`logs/module-1/log-stat ements-catalogue.csv:1`
					`code/module-1/fronten d.py:104`
					high 90%

FR-007	**Refresh balance cache** System updates the cached balance information	system Balance update or cache expiration	account identifier, new balance amount → updated balance cache	• Cache balances for performance • Update cache when balances change	`telemetrics/module-1/t elemetry-signals-catal ogue.csv:1` high 90%

Payments and Transfer

ID	Requirement	Actor / trigger	Inputs → outputs	Business rules	Source

FR-008	**Make payment**
	Customer transfers funds to another account
	`POST /payment`
		customer
		User submits payment form
			destination account number, destination routing number, amount, authentication token → transaction confirmation, transaction UUID	• Payments must be numeric and cannot be submitted twice
				• Customers can overdraw accounts by up to $500 on payments
				• Each transaction has a unique UUID to prevent duplicates
					`code/module-1/frontend.py:240`
					`logs/module-1/log-statements-catalogue.csv:1`
					high 90%

FR-009	**View transaction history**
	Customer views their past transaction records
	`GET /transactions/{account_id}`
		customer
		User requests transaction history
			account identifier, authentication token → list of transactions with details	• User authentication is required
				• Transaction data is cached for performance
					`telemetrics/module-1/telemetry-signals-catalogue.csv:1`
					`code/module-1/frontend.py:104`
					high 90%

FR-010	**Validate transaction**
	System validates a payment transaction before processing
		system
		Payment submission request
			transaction details, account information → validation result, error messages if invalid	• Payments must be numeric
				• Account numbers must be valid
				• Routing numbers must be valid
					`code/module-1/frontend.py:257`
					`brd/module-1/bank-of-anthos-BRD-draft-v0.4.md:FR-11`
					high 90%

FR-011	**Process transaction**
	System processes a validated payment transaction
		system
		Validated transaction submission
			validated transaction details → transaction record, updated balances	• Update both source and destination accounts
				• Record transaction with timestamp and UUID
					`logs/module-1/log-statements-catalogue.csv:1`
					`telemetrics/module-1/telemetry-signals-catalogue.csv:1`
					high 90%

Deposits

ID	Requirement	Actor / trigger	Inputs → outputs	Business rules	Source

FR-012	**Make deposit**
	Customer deposits funds from an external account
	`POST /deposit`
		customer
		User submits deposit form
			external account number, external routing number, amount, authentication token → deposit confirmation, deposit UUID	• External deposits cannot use the local routing number
				• Each deposit has a unique UUID to prevent duplicates
					`code/module-1/frontend.py:307`
					`code/module-1/frontend.py:278`
					high 90%

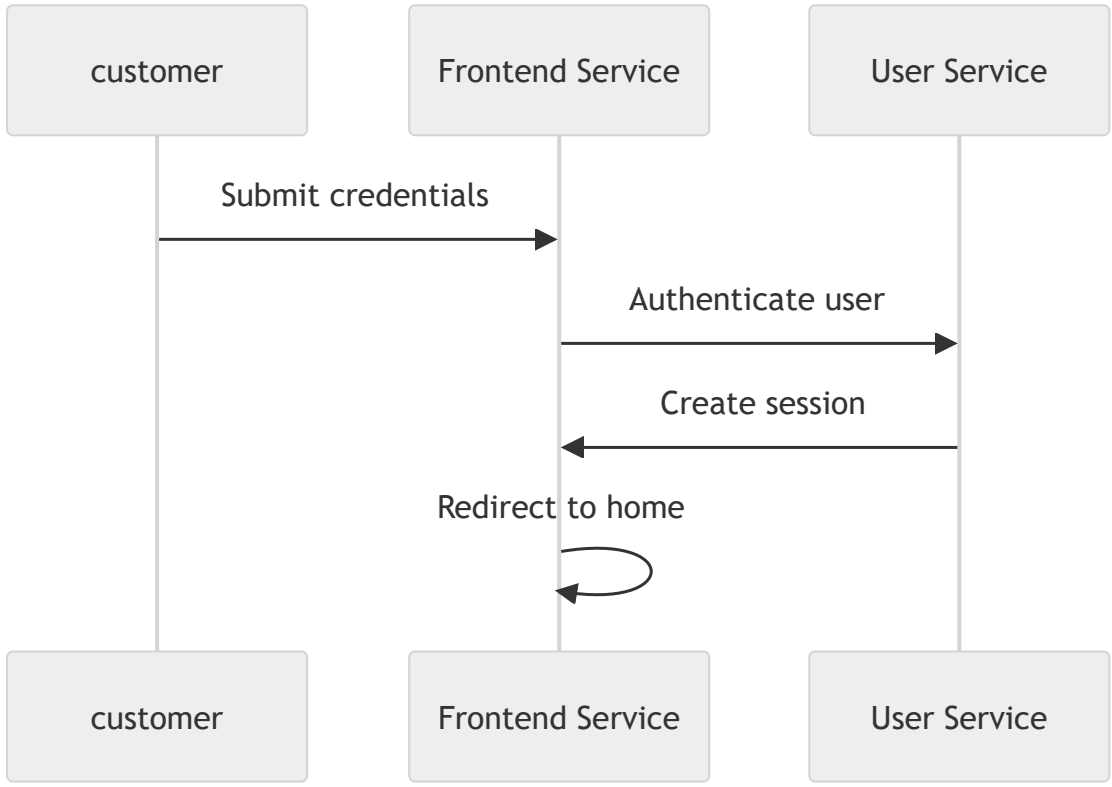
8 Execution & business flows

User Login Flow

Overview · actor: customer · starts at: scr-login · high 90%

End-to-end sequence for customer authentication

1. **Submit credentials** Frontend Service — Customer enters username and password on login screen
2. **Authenticate user** User Service — User service validates credentials and generates JWT token Reads: User.
3. **Create session** Frontend Service — Frontend stores authentication token for session
4. **Redirect to home** Frontend Service — System redirects authenticated user to home screen



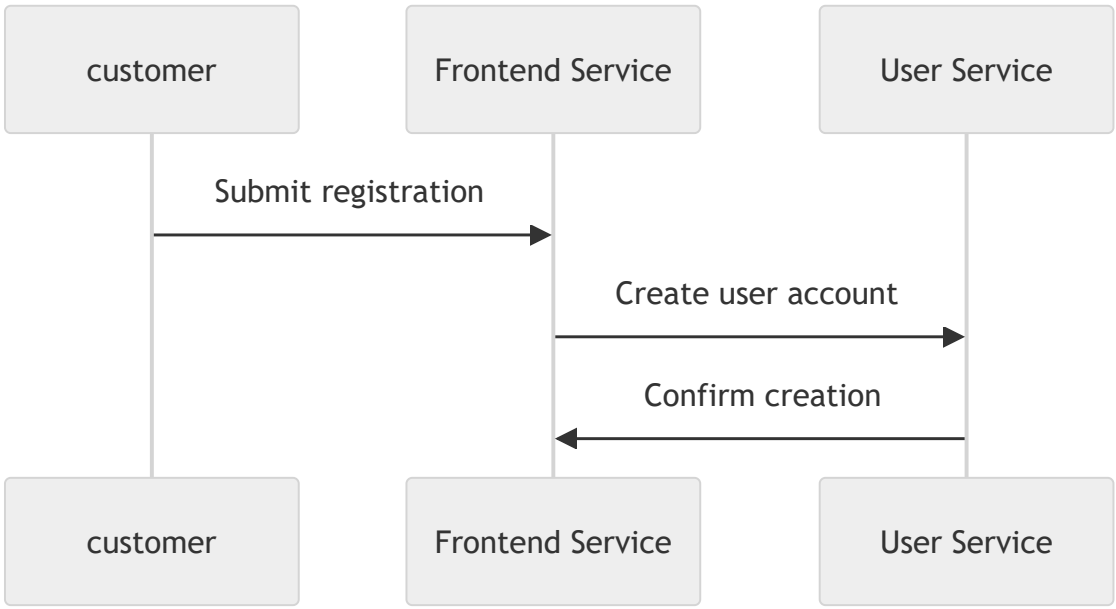
Sequence: User Login Flow

User Signup Flow

Overview · actor: customer · starts at: scr-signup · high 90%

End-to-end sequence for new customer account creation

1. **Submit registration** Frontend Service — Customer enters personal information and credentials
2. **Create user account** User Service — User service validates and creates new user record Writes: User.
3. **Confirm creation** Frontend Service — Frontend displays confirmation to customer



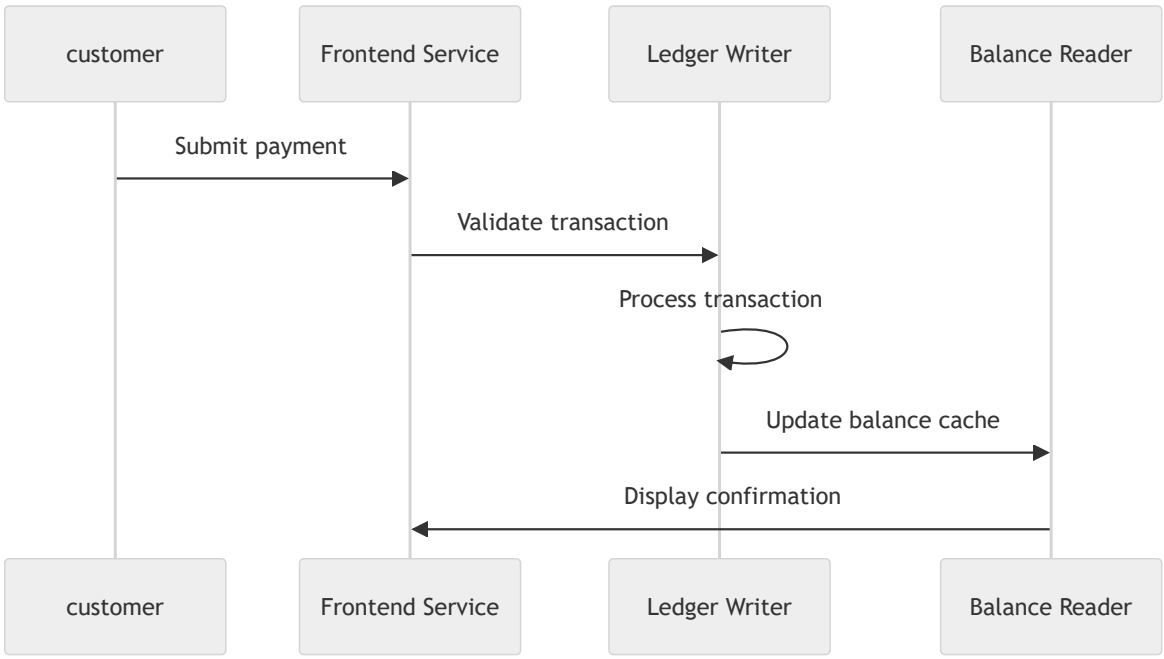
Sequence: User Signup Flow

Make Payment Flow

Payments and Transfer · actor: customer · starts at: scr-home · high 90%

End-to-end sequence for transferring funds between accounts

1. **Submit payment** Frontend Service — Customer enters payment details on home screen
2. **Validate transaction** Ledger Writer — System validates payment details and account information Reads: Account, Balance.
3. **Process transaction** Ledger Writer — System processes validated payment and updates balances Reads: Account, Balance. Writes: Transaction, Balance.
4. **Update balance cache** Balance Reader — System refreshes cached balance information Reads: Balance.
5. **Display confirmation** Frontend Service — Frontend shows payment confirmation to customer Reads: Transaction.



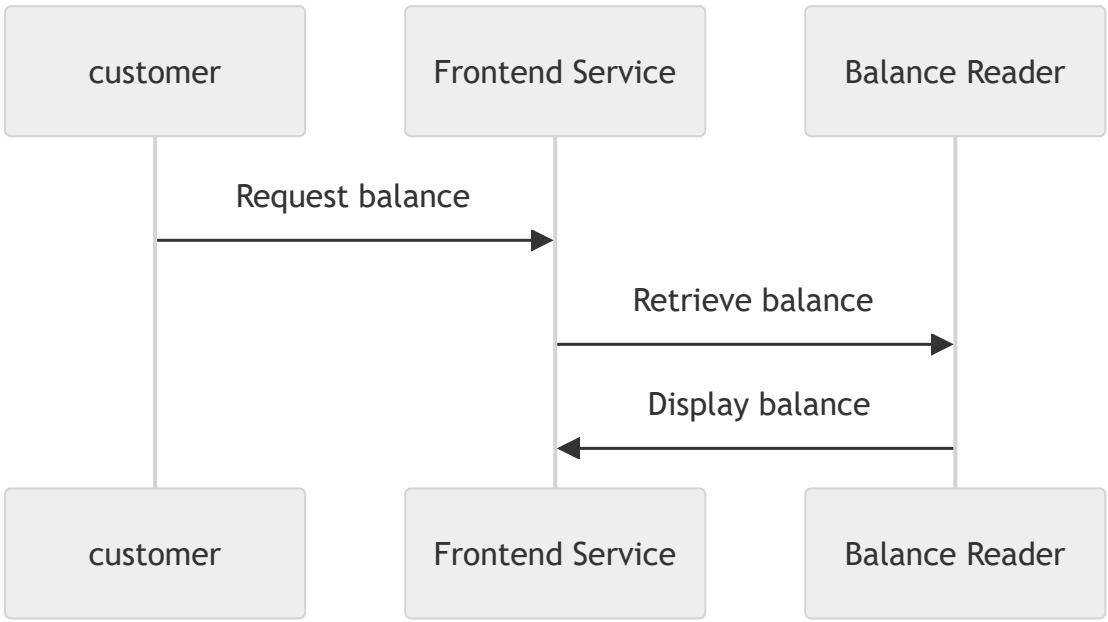
Sequence: Make Payment Flow

View Balance Flow

Accounts · actor: customer · starts at: scr-home · high 90%

End-to-end sequence for retrieving account balance

1. **Request balance** Frontend Service — Customer navigates to home screen to view balance
2. **Retrieve balance** Balance Reader — Balance reader fetches current balance from database Reads: Balance.
3. **Display balance** Frontend Service — Frontend shows balance to customer Reads: Balance.



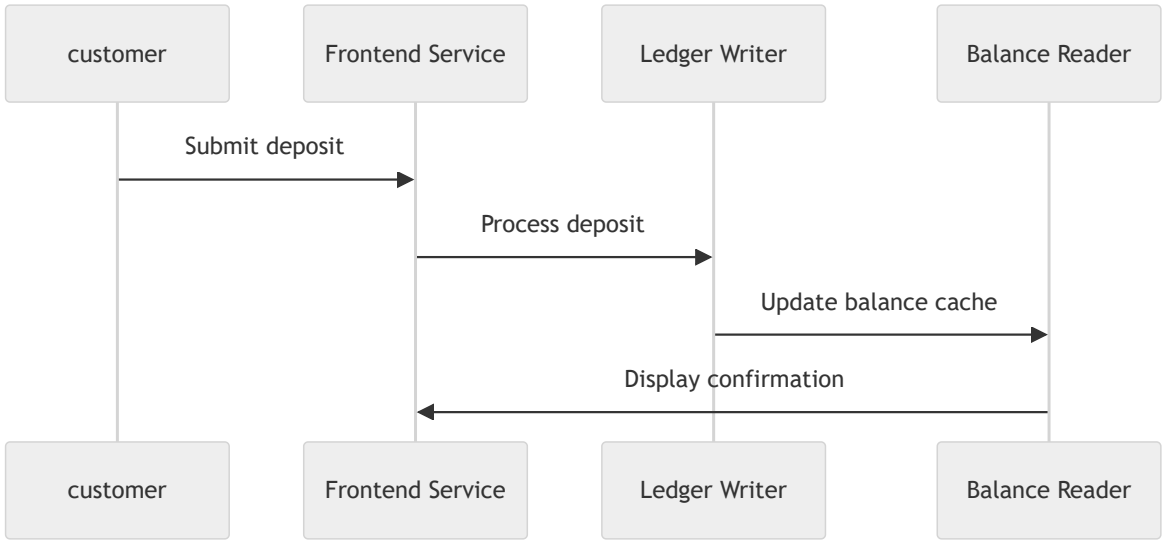
Sequence: View Balance Flow

Make Deposit Flow

Deposits · actor: customer · starts at: scr-home · high 90%

End-to-end sequence for depositing funds from external accounts

1. **Submit deposit** Frontend Service — Customer enters deposit details on home screen
2. **Process deposit** Ledger Writer — System processes deposit and updates account balance Reads: Account, Balance. Writes: Deposit, Balance.
3. **Update balance cache** Balance Reader — System refreshes cached balance information Reads: Balance.
4. **Display confirmation** Frontend Service — Frontend shows deposit confirmation to customer Reads: Deposit.



Sequence: Make Deposit Flow

9 Personas & user journeys

John Customer — Bank Customer

Goals

- Check account balance regularly
- Send payments to other accounts
- Deposit funds from external accounts
- Manage payment contacts
- View transaction history

Pain points

- Limited session duration (15 minutes)
- No multi-factor authentication
- No mobile app access
- Limited payment options
- No overdraft protection beyond $500

Journey: Customer Login Journey

Customer authentication journey to access banking services

DISCOVER	ONBOARD	TRANSACT	REVIEW
Customer navigates to bank website *via Web browser* ⚠ No mobile app option	Customer accesses login screen *via Login screen* ⚠ No password reset option	Customer enters credentials and authenticates *via Authentication form* ⚠ Session expires after 15 minutes	Customer views account dashboard *via Home screen* ⚠ No transaction categorization

Journey: Customer Payment Journey

Customer payment journey to transfer funds between accounts

DISCOVER	ONBOARD	TRANSACT	REVIEW
Customer navigates to payment section *via Home screen* ⚠ Limited payment options	Customer selects payment recipient *via Contact list* ⚠ No saved payment templates	Customer enters payment amount and submits *via Payment form* ⚠ No overdraft protection beyond $500	Customer views payment confirmation *via Transaction history* ⚠ No immediate notification

Bank Operator — Bank Operations Staff

Goals

- Monitor system health and performance
- Investigate failed transactions
- Manage customer accounts
- Process exceptions
- Generate reports for management

Pain points

- Limited monitoring dashboard
- No automated alerting system
- No comprehensive audit logs
- Limited reporting capabilities
- No fraud detection system

Journey: Operator Monitoring Journey

Bank operator journey to monitor system health and performance

DISCOVER
Operator accesses monitoring dashboard
via System health dashboard
⚠ Limited metrics available

ONBOARD
Operator reviews system status
via Dashboard widgets
⚠ No automated alerting

TRANSACT
Operator investigates failed transactions
via Transaction logs
⚠ No comprehensive audit trail

REVIEW
Operator generates performance report
via Report generator
⚠ Limited reporting capabilities

Dev Engineer — System Developer

Goals

- Maintain and update banking services
- Ensure system scalability
- Implement new features
- Debug performance issues
- Monitor service dependencies

Pain points

- Limited observability tools
- No comprehensive testing framework
- No automated deployment pipeline
- Limited documentation
- No performance SLAs

Journey: Engineer Maintenance Journey

Engineer journey to maintain and update banking services

DISCOVER
Engineer reviews service health
via Kubernetes dashboard
⚠ Limited observability tools

ONBOARD
Engineer checks service dependencies
via Service registry
⚠ No dependency mapping

TRANSACT
Engineer deploys service updates
via CI/CD pipeline
⚠ No automated testing

REVIEW
Engineer monitors post-deployment performance
via Telemetry dashboard
⚠ No performance SLAs

10 Reports

Report	Module	Status	Audience	Key fields	Frequency
Account Balance Report Shows current balance for a specific account	Accounts	Exists today	customer	account number, current balance, last updated timestamp	on demand
Transaction History Report Shows historical transactions for an account	Payments and Transfer	Exists today	customer	transaction ID, date, amount, from account, to account, status	on demand
Monthly Statement Comprehensive monthly account statement with transactions and balances	Accounts	Required — not built	customer, regulators	account details, monthly transactions, opening/closing balances, fees, interest	monthly
Account Summary Dashboard Overview of all customer accounts with balances and recent activity	Accounts	Exists today	customer, customer service	account numbers, current balances, recent transactions, contact information	on demand
Payment Failures Report Tracks failed payment transactions for investigation	Payments and Transfer	Required — not built	operations, fraud detection	transaction ID, amount, timestamp, failure reason, account numbers	daily
Deposit Summary Report Summary of external deposits into customer accounts	Deposits	Exists today	customer, operations	deposit ID, amount, timestamp, external account, destination account	on demand
Contact List Report Lists all payment contacts for a customer	Accounts	Exists today	customer	contact name, account number, routing number, external flag	on demand
Fraud Detection Report Identifies potentially fraudulent transactions for investigation	Payments and Transfer	Required — not built	fraud detection, security	suspicious transactions, risk scores, account patterns, alert details	real-time
Regulatory Compliance Report Ensures banking operations meet regulatory requirements	Overview	Required — not built	compliance, regulators	compliance metrics, audit trails, policy violations, remediation actions	monthly
System Health Dashboard Monitors overall system performance and availability	Overview	Exists today	operations, devops	service status, response times, error rates, resource utilization	real-time
User Activity Report Tracks user login and activity patterns	Overview	Required — not built	security, customer service	user ID, login times, IP addresses, failed attempts, session duration	daily
Card Activity Report Tracks credit/debit card transactions and usage	Cards	Required — not built	customer, fraud detection	card number, transactions, locations, amounts, flags	daily
Loan Status Report Shows loan application status and payment history	Loans	Required — not built	customer, loan officers	loan application, status, amount, term, payments, balance	on demand
Account Reconciliation Report Reconciles account balances with transaction records	Accounts	Required — not built	operations, auditors	account balance, transaction totals, discrepancies, adjustments	daily

Report	Module	Status	Audience	Key fields	Frequency
Audit Log Comprehensive log of system changes and administrative actions	Overview	Required — not built	auditors, security	timestamp, user, action, resource, old value, new value	real-time

11 Capabilities, epics & user stories

Overview full

Capability	Status	Description	Realised by
User Authentication	implemented	Ability to authenticate users with credentials and manage sessions	User login, User signup, User logout
Regulatory Compliance	gap	Ability to ensure operations meet regulatory requirements	—
Audit Logging	gap	Ability to maintain comprehensive logs of system changes	—

Epic · Identity Management Enhancement

Improve user authentication and session management capabilities

Feature · Multi-Factor Authentication

STORY-001 Enroll in Multi-Factor Authentication — must

As a **a bank customer**, I want **to enroll in multi-factor authentication for my account**, so that **I can add an extra layer of security to my login process**.

☐ Given I am logged into my account, when I navigate to security settings, then I should see an option to enroll in multi-factor authentication
☐ Given I choose to enroll in MFA, when I provide my phone number, then I should receive a verification code
☐ Given I receive the verification code, when I enter it correctly, then my account should be enrolled in MFA
☐ Given I am enrolled in MFA, when I log in, then I should be prompted to enter the verification code

STORY-002 Login with Multi-Factor Authentication — must

As a **a bank customer**, I want **to log in using multi-factor authentication**, so that **I can securely access my account with enhanced security**.

☐ Given I am on the login page, when I enter my username and password, then I should be prompted for my verification code
☐ Given I am prompted for verification code, when I enter the correct code, then I should be logged in successfully
☐ Given I enter an incorrect verification code, when I attempt to login, then I should see an error message and be allowed to try again
☐ Given I fail to enter the correct code multiple times, then my account should be temporarily locked for security

STORY-003 Recover Multi-Factor Authentication — should

As a **a bank customer**, I want **to recover access to my account if I lose my authentication device**, so that **I can still access my account in case of lost or broken devices**.

☐ Given I am locked out of my account due to lost authentication device, when I contact customer support, then I should be able to verify my identity
☐ Given I have verified my identity, when I request MFA recovery, then I should be able to enroll a new authentication device
☐ Given I enroll a new device, when I attempt to login, then I should be prompted with the new authentication method
☐ Given I have recovered my account, then I should receive a notification about the recovery event

Feature · Enhanced Session Management

STORY-004 Remember Device Option — could

As a **a bank customer**, I want **to remember my login device**, so that **I don't need to authenticate as frequently on trusted devices**.

- ☐ Given I am on the login page, when I check 'remember this device', then my session should persist longer on that device
- ☐ Given I have chosen to remember my device, when I return to the site, then I should be automatically logged in
- ☐ Given I am on a remembered device, when I attempt to perform sensitive actions, then I may still be asked to re-authenticate
- ☐ Given I want to remove a remembered device, when I go to security settings, then I should see an option to remove trusted devices

STORY-005 Session Activity Monitoring — should

As a **a bank customer**, I want **to view my recent account activity**, so that **I can monitor account access and detect unauthorized activity**.

- ☐ Given I am logged in, when I navigate to security settings, then I should see a list of recent login sessions
- ☐ Given I see a suspicious session, when I select it, then I should see details including location and device information
- ☐ Given I identify an unauthorized session, when I choose to end it, then that session should be immediately terminated
- ☐ Given I end a session, then I should receive a confirmation that the session has been ended

Feature · Comprehensive Audit Logging

STORY-006 View Account Activity Log — should

As a **a bank customer**, I want **to view a detailed log of my account activities**, so that **I can monitor all actions performed on my account**.

- ☐ Given I am logged in, when I navigate to account settings, then I should see an option to view my activity log
- ☐ Given I view my activity log, then I should see a chronological list of all actions performed on my account
- ☐ Given I view my activity log, then I should be able to filter by date range or action type
- ☐ Given I see an activity I don't recognize, when I select it, then I should see detailed information about that action

STORY-007 Administrative Audit Access — must

As a **a bank operator**, I want **to access detailed audit logs for customer accounts**, so that **I can investigate issues and ensure compliance**.

- ☐ Given I am a bank operator, when I search for a customer account, then I should have access to their audit log
- ☐ Given I access a customer's audit log, then I should see all actions performed on that account
- ☐ Given I see suspicious activity, when I flag it, then it should be escalated to the security team
- ☐ Given I access an audit log, then my access should be logged for compliance purposes

Accounts partial

Capability	Status	Description	Realised by
Account Management	implemented	Ability to create and manage customer accounts with personal information	View account balance, Add contact
Balance Management	implemented	Ability to retrieve and update account balances with caching	View account balance, Refresh balance cache
Contact Management	implemented	Ability to manage payment contacts for customers	Add contact

Epic · Account Management Enhancement

Improve account management capabilities and reporting

Feature · Enhanced Account Summary

STORY-008 View Account Overview — must

As a **a bank customer**, I want **to see a comprehensive overview of all my accounts**, so that **I can quickly understand my financial situation across all accounts**.

- ☐ Given I am logged in, when I navigate to the home page, then I should see all my accounts with their current balances
- ☐ Given I see my accounts, when I select an account, then I should see recent transactions for that account
- ☐ Given I have multiple accounts, when I view the overview, then I should see the total balance across all accounts
- ☐ Given I view my accounts, then I should be able to filter by account type or status

STORY-009 View Detailed Account Information — should

As a **a bank customer**, I want **to view detailed information about a specific account**, so that **I can see account details, terms, and conditions**.

- ☐ Given I am on the account overview page, when I select an account, then I should see detailed account information
- ☐ Given I view account details, then I should see account number, routing number, opening date, and interest rate
- ☐ Given I view account details, then I should see account terms and conditions
- ☐ Given I view account details, then I should see options to perform account actions like close or rename

Feature · Monthly Statement Generation

STORY-010 Generate Monthly Statement — must

As a **a bank customer**, I want **to receive a monthly statement of my account activity**, so that **I can track my financial activities and reconcile my records**.

- ☐ Given it is the end of the month, when the system processes statements, then I should receive a statement for each of my accounts
- ☐ Given I receive a statement, then it should include all transactions for the month with dates and amounts
- ☐ Given I receive a statement, then it should show opening and closing balances for the month
- ☐ Given I receive a statement, then I should be able to download it as a PDF file

STORY-011 Statement Delivery Options — should

As a **a bank customer**, I want **to choose how I receive my monthly statements**, so that **I can receive statements in my preferred format**.

- ☐ Given I am in account settings, when I navigate to statement preferences, then I should see options for delivery method
- ☐ Given I choose electronic delivery, then I should receive my statement via email
- ☐ Given I choose paper delivery, then I should receive a physical statement in the mail
- ☐ Given I change my delivery preference, then my future statements should be delivered according to the new preference

Feature · Enhanced Contact Management

STORY-012 Organize Contacts into Groups — should

As a **a bank customer**, I want **to organize my contacts into groups**, so that **I can manage my contacts more efficiently**.

- ☐ Given I am on the contacts page, when I view my contacts, then I should see an option to create groups
- ☐ Given I create a group, when I add contacts to it, then those contacts should appear in the group
- ☐ Given I have multiple groups, when I select a group, then I should only see contacts in that group
- ☐ Given I have contacts in groups, when I make a payment, then I should be able to select from groups

STORY-013 Create Payment Templates — could

As a **a bank customer**, I want **to create payment templates for frequent transfers**, so that **I can make recurring payments more quickly**.

- ☐ Given I am on the contacts page, when I select a contact, then I should see an option to create a payment template
- ☐ Given I create a payment template, when I set the amount and frequency, then it should be saved
- ☐ Given I have payment templates, when I navigate to payments, then I should see my templates
- ☐ Given I select a template, when I initiate a payment, then the details should be pre-filled

Payments and Transfer — partial

Capability	Status	Description	Realised by
Payment Processing	implemented	Ability to process payments between accounts with validation	Make payment, Validate transaction, Process transaction
Transaction History	implemented	Ability to retrieve and display historical transactions	View transaction history
Fraud Detection	gap	Ability to detect and flag potentially fraudulent transactions	—

Epic · Payment Processing Enhancement

Improve payment processing capabilities and fraud detection

Feature · Scheduled and Recurring Payments

STORY-014 Schedule a Future Payment — should

As a **a bank customer**, I want **to schedule a payment for a future date**, so that **I can plan my payments in advance**.

- ☐ Given I am on the payment page, when I choose to schedule a payment, then I should be able to select a future date
- ☐ Given I select a future date, when I submit the payment, then it should be scheduled for that date
- ☐ Given I have a scheduled payment, when I view my upcoming payments, then I should see it listed
- ☐ Given I have a scheduled payment, when I want to cancel it, then I should be able to cancel before the scheduled date

STORY-015 Create a Recurring Payment — should

As a **a bank customer**, I want **to create a recurring payment**, so that **I can make regular payments automatically**.

- ☐ Given I am on the payment page, when I choose to create a recurring payment, then I should be able to set frequency
- ☐ Given I set the frequency, when I configure the payment, then it should repeat according to the schedule
- ☐ Given I have a recurring payment, when I view my upcoming payments, then I should see all upcoming instances
- ☐ Given I have a recurring payment, when I want to modify it, then I should be able to change the amount or frequency

Feature · Fraud Detection System

STORY-016 Monitor for Suspicious Activity — must

As a **a bank fraud analyst**, I want **to monitor transactions for suspicious patterns**, so that **I can detect and prevent fraudulent activity**.

- ☐ Given I am a fraud analyst, when I access the fraud dashboard, then I should see flagged transactions
- ☐ Given I see a flagged transaction, when I select it, then I should see details about why it was flagged
- ☐ Given I investigate a transaction, when I determine it's fraudulent, then I should be able to block the transaction
- ☐ Given I block a transaction, then the customer should be notified and the transaction reversed

STORY-017 Customer Fraud Alerts `must`

As a **a bank customer**, I want **to receive alerts about suspicious activity on my account**, so that **I can quickly respond to potential fraud**.

- ☐ Given there is suspicious activity on my account, when it is detected, then I should receive an immediate alert
- ☐ Given I receive a fraud alert, when I view it, then I should see details about the suspicious activity
- ☐ Given I receive a fraud alert, when I confirm it's fraudulent, then the transaction should be blocked
- ☐ Given I receive a fraud alert, when I confirm it's legitimate, then my account should be unlocked

Feature · Enhanced Payment Validation

STORY-018 Configure Payment Validation Rules `should`

As a **a bank operator**, I want **to configure validation rules for payments**, so that **I can enforce business rules and prevent errors**.

- ☐ Given I am a bank operator, when I access the payment settings, then I should see validation rules configuration
- ☐ Given I configure validation rules, when a payment is processed, then it should be validated against those rules
- ☐ Given a payment fails validation, when the customer attempts it, then they should receive a clear error message
- ☐ Given I configure validation rules, when I save them, then they should be immediately active

STORY-019 Real-Time Payment Validation `should`

As a **a bank customer**, I want **to receive immediate feedback on payment validation**, so that **I can correct errors before submitting**.

- ☐ Given I am entering payment details, when I enter an invalid account number, then I should see an immediate error
- ☐ Given I am entering payment details, when I enter an amount that exceeds my balance, then I should see a warning
- ☐ Given I am entering payment details, when I enter a duplicate payment, then I should be notified
- ☐ Given I correct an error, when I resubmit, then the payment should proceed normally

Deposits `partial`

Capability	Status	Description	Realised by
Deposit Processing	implemented	Ability to process deposits from external accounts	Make deposit

Epic · Deposit Processing Enhancement

Improve deposit processing capabilities and reporting

Feature · Enhanced Deposit Tracking

STORY-020 View Deposit Status `must`

As a **a bank customer**, I want **to see the status of my deposits**, so that **I can know when my funds will be available**.

- ☐ Given I submit a deposit, when I view my transaction history, then I should see the deposit status
- ☐ Given I have a pending deposit, when I check its status, then I should see estimated completion time
- ☐ Given I have a completed deposit, when I view it, then I should see confirmation and availability time
- ☐ Given I have a failed deposit, when I view it, then I should see the reason for failure

STORY-021 View Deposit History `should`

As a **a bank customer**, I want **to view my deposit history**, so that **I can track all my deposits over time**.

- ☐ Given I am logged in, when I navigate to deposit history, then I should see all my past deposits
- ☐ Given I view my deposit history, then I should be able to filter by date range
- ☐ Given I view my deposit history, then I should be able to search by deposit amount or source
- ☐ Given I select a deposit from history, then I should see detailed information about that deposit

Feature · Deposit Verification System

STORY-022 Verify External Deposits — must

As a **a bank operator**, I want **to verify large or unusual deposits**, so that **I can ensure compliance and prevent fraud**.

- ☐ Given I am a bank operator, when I access the deposit verification queue, then I should see deposits requiring verification
- ☐ Given I select a deposit for verification, when I review it, then I should see all relevant details
- ☐ Given I verify a deposit, when I approve it, then the funds should be credited to the account
- ☐ Given I verify a deposit, when I reject it, then the customer should be notified

STORY-023 Configure Deposit Automation Rules — should

As a **a bank operator**, I want **to configure rules for automatic deposit verification**, so that **I can streamline the verification process**.

- ☐ Given I am a bank operator, when I access deposit settings, then I should see automation rules configuration
- ☐ Given I configure automation rules, when a deposit matches the criteria, then it should be automatically verified
- ☐ Given I configure automation rules, when a deposit doesn't match, then it should be sent for manual verification
- ☐ Given I configure automation rules, when I save them, then they should be immediately active

Feature · Deposit Reporting and Analytics

STORY-024 View Deposit Summary Report — should

As a **a bank customer**, I want **to see a summary of my deposits**, so that **I can track my deposit activity**.

- ☐ Given I am logged in, when I navigate to reports, then I should see a deposit summary option
- ☐ Given I view the deposit summary, then I should see total deposits by month
- ☐ Given I view the deposit summary, then I should see deposits by source account
- ☐ Given I view the deposit summary, then I should be able to export the data

STORY-025 Deposit Analytics Dashboard — should

As a **a bank operator**, I want **to access deposit analytics**, so that **I can identify trends and issues**.

- ☐ Given I am a bank operator, when I access the analytics dashboard, then I should see deposit metrics
- ☐ Given I view the analytics dashboard, then I should see deposit volume trends over time
- ☐ Given I view the analytics dashboard, then I should see average deposit amounts
- ☐ Given I view the analytics dashboard, then I should see any unusual deposit patterns

Cards none

Capability	Status	Description	Realised by
Card Management	gap	Ability to issue, manage, and monitor credit/debit cards	—

Epic · Card Management System

Implement comprehensive card management capabilities

Feature · Card Issuance

STORY-026 Apply for a New Card — must

As a **a bank customer**, I want **to apply for a new credit or debit card**, so that **I can have access to additional payment methods**.

- ☐ Given I am logged in, when I navigate to card services, then I should see an option to apply for a new card
- ☐ Given I choose to apply for a card, when I complete the application, then I should receive confirmation
- ☐ Given I submit an application, when it's approved, then I should receive the card in the mail
- ☐ Given I submit an application, when it's declined, then I should receive notification with reasons

STORY-027 Select Card Type `should`

As a **a bank customer**, I want **to choose from different card types**, so that **I can select the card that best fits my needs**.

- ☐ Given I am applying for a card, when I view available options, then I should see different card types
- ☐ Given I select a card type, when I view details, then I should see features and benefits
- ☐ Given I select a card type, when I apply, then the application should be for that specific type
- ☐ Given I apply for a card, then I should be able to see the status of my application

Feature · Card Management Features

STORY-028 Activate a New Card `must`

As a **a bank customer**, I want **to activate my new card**, so that **I can start using it**.

- ☐ Given I receive a new card, when I log in to activate it, then I should see an activation option
- ☐ Given I choose to activate my card, when I enter the required information, then it should be activated
- ☐ Given I activate my card, then I should receive confirmation
- ☐ Given I activate my card, then I should be able to use it immediately

STORY-029 Manage Card Controls `should`

As a **a bank customer**, I want **to set controls and limits on my cards**, so that **I can manage my card usage and security**.

- ☐ Given I am logged in, when I navigate to card services, then I should see card controls
- ☐ Given I view card controls, then I should be able to set spending limits
- ☐ Given I view card controls, then I should be able to enable or disable online purchases
- ☐ Given I change card controls, then they should be immediately effective

Feature · Card Activity Monitoring

STORY-030 View Card Transactions `must`

As a **a bank customer**, I want **to view my card transaction history**, so that **I can track my spending**.

- ☐ Given I am logged in, when I navigate to card services, then I should see my cards
- ☐ Given I select a card, when I view transactions, then I should see recent activity
- ☐ Given I view transactions, then I should be able to filter by date range
- ☐ Given I view transactions, then I should be able to search by merchant or amount

STORY-031 Set Card Transaction Alerts `should`

As a **a bank customer**, I want **to receive alerts for card transactions**, so that **I can monitor my card activity**.

- ☐ Given I am logged in, when I navigate to card alerts, then I should see alert options
- ☐ Given I set an alert, when a transaction occurs, then I should receive a notification
- ☐ Given I set an alert, when a transaction exceeds my threshold, then I should receive an immediate alert
- ☐ Given I set an alert, then I should be able to customize the alert conditions

Loans `none`

Capability	Status	Description	Realised by
Loan Processing	gap	Ability to originate, service, and manage loans	—

Epic · Loan Processing System

Implement comprehensive loan origination and servicing capabilities

Feature · Loan Origination

STORY-032 Apply for a Loan — must

As a **a bank customer**, I want **to apply for a loan**, so that **I can access funds for major purchases**.

- ☐ Given I am logged in, when I navigate to loan services, then I should see loan options
- ☐ Given I select a loan type, when I start the application, then I should see the required information
- ☐ Given I complete the application, when I submit it, then I should receive confirmation
- ☐ Given I submit an application, when it's under review, then I should be able to check the status

STORY-033 Prequalify for a Loan — should

As a **a bank customer**, I want **to prequalify for a loan**, so that **I can understand my borrowing options**.

- ☐ Given I am logged in, when I navigate to loan services, then I should see a prequalification option
- ☐ Given I start prequalification, when I provide my information, then I should see potential loan options
- ☐ Given I view prequalification results, then I should see estimated loan amounts and terms
- ☐ Given I prequalify, then I should be able to proceed to a full application

Feature · Loan Servicing

STORY-034 View Loan Dashboard — must

As a **a bank customer**, I want **to see an overview of my loans**, so that **I can manage my loan accounts**.

- ☐ Given I am logged in, when I navigate to loan services, then I should see my loans
- ☐ Given I view my loans, then I should see current balances and payment status
- ☐ Given I view my loans, then I should see upcoming payment dates
- ☐ Given I view my loans, then I should be able to see payment history

STORY-035 Make Loan Payments — must

As a **a bank customer**, I want **to make payments on my loans**, so that **I can stay current on my obligations**.

- ☐ Given I am on my loan dashboard, when I select a loan, then I should see payment options
- ☐ Given I choose to make a payment, when I enter the amount, then I should be able to complete the transaction
- ☐ Given I make a payment, then I should receive confirmation
- ☐ Given I make a payment, then my loan balance should be updated

Feature · Loan Analytics and Reporting

STORY-036 Generate Loan Reports — should

As a **a bank loan officer**, I want **to generate reports on loan performance**, so that **I can assess portfolio health**.

- ☐ Given I am a loan officer, when I access the reporting dashboard, then I should see loan report options
- ☐ Given I generate a loan report, then I should see portfolio metrics
- ☐ Given I generate a loan report, then I should see delinquency rates
- ☐ Given I generate a loan report, then I should be able to export the data

STORY-037 Loan Analytics Dashboard — should

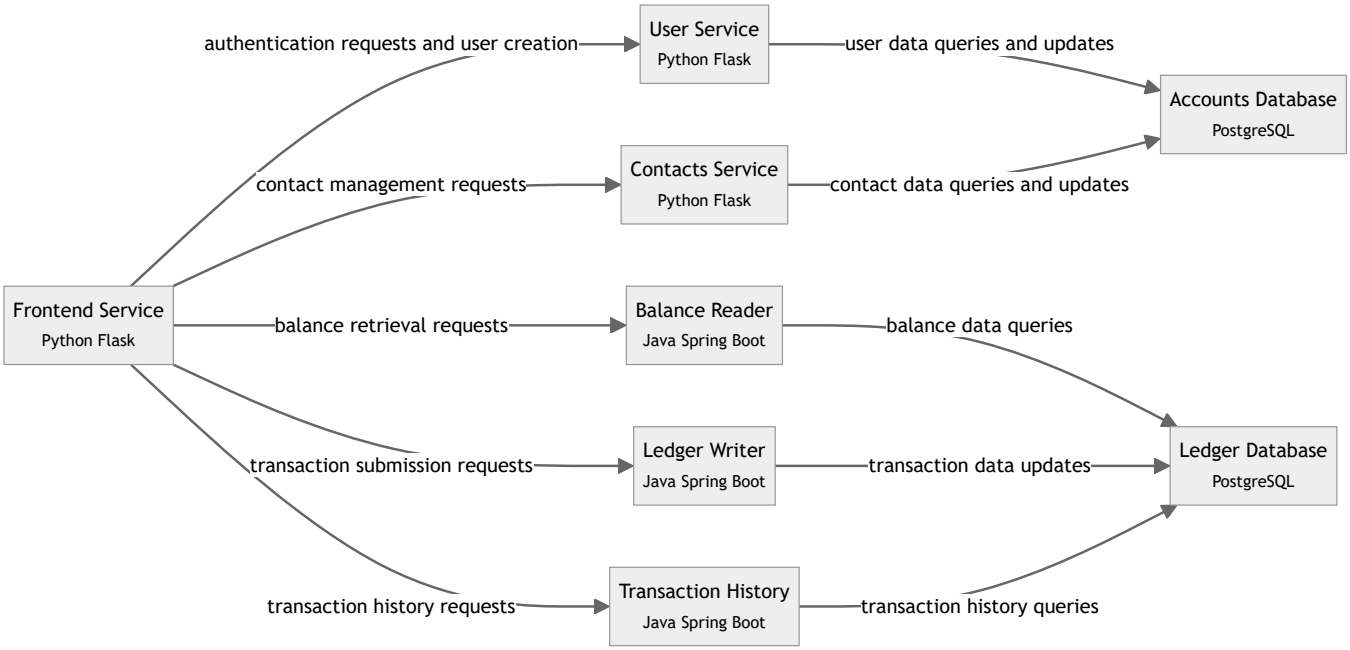
As a **a bank analyst**, I want **to access loan analytics**, so that **I can identify trends and opportunities**.

- ☐ Given I am a bank analyst, when I access the analytics dashboard, then I should see loan metrics
- ☐ Given I view the analytics dashboard, then I should see loan volume trends
- ☐ Given I view the analytics dashboard, then I should see default rate trends
- ☐ Given I view the analytics dashboard, then I should see performance by loan type

12 Business rules

ID	Rule	Module	Source
BR-AUTH-SESSION	User sessions expire after 15 minutes of inactivity	Overview	`brd/module-1/bank-of-anthos-BRD-draft-v0.4.md:FR-04`
BR-AUTH-TOKEN	JWT tokens are used for authentication and authorization	Overview	`code/module-1/frontend.py:625` `code/module-1/userservice.py:199`
BR-AUTH-OAUTH	Third-party applications require explicit user consent	Overview	`code/module-1/frontend.py:527` `brd/module-1/bank-of-anthos-BRD-draft-v0.4.md:FR-05`
BR-PAYMENT-VALIDATION	Payments must be numeric and cannot be submitted twice	Payments and Transfer	`code/module-1/frontend.py:257` `brd/module-1/bank-of-anthos-BRD-draft-v0.4.md:FR-11`
BR-OVERDRAFT	Customers can overdraw accounts by up to $500 on payments	Payments and Transfer	`brd/module-1/bank-of-anthos-BRD-draft-v0.4.md:FR-10`
BR-DEPOSIT-VALIDATION	External deposits cannot use the local routing number	Deposits	`code/module-1/frontend.py:278`
BR-USER-CREATION	Usernames must be 2-15 alphanumeric characters or underscores	Overview	`code/module-1/userservice.py:135`
BR-PASSWORD-SECURITY	Passwords are stored as bcrypt hashes with salt	Overview	`code/module-1/userservice.py:96`
BR-ACCOUNT-NUMBERS	Account numbers are 10-digit unique identifiers	Accounts	`code/module-1/userservice.py:69` `database/module-1/accounts-demo-data.sql:1`
BR-ROUTING-NUMBERS	Routing numbers are 9-digit identifiers for banks	Payments and Transfer	`code/module-1/frontend.py:240` `database/module-1/accounts-demo-data.sql:1`
BR-TRANSACTION-ID	Each transaction has a unique UUID to prevent duplicates	Payments and Transfer	`code/module-1/frontend.py:330`
BR-CONTACT-VALIDATION	Contacts must have unique labels per user	Accounts	`logs/module-1/log-statements-catalogue.csv:1`

13 Architecture, integrations & non-functional requirements



Containers and their interactions

Integrations

Integration	Counterpart	Direction	Protocol	Data exchanged
Google Cloud Metadata Server	Google Cloud Platform	outbound	HTTP	Cluster name, pod name, zone information
Google Cloud Trace	Google Cloud Platform	outbound	HTTP	Trace spans and metadata
Google Cloud Monitoring	Google Cloud Platform	outbound	HTTP	Service metrics and health data
Google Cloud Logging	Google Cloud Platform	outbound	HTTP	Application logs and events
OAuth Provider	OAuth Provider	inbound	HTTP	Authorization codes and tokens
External Bank Accounts	External Banks	inbound	HTTP	Deposit funds and routing information

Deployment

Unit	Runtime	Scale
Accounts Database	PostgreSQL	1 replica
Balance Reader Service	Java Spring Boot	1 replica
Contacts Service	Python Flask	1 replica
Frontend Service	Python Flask	1 replica
Ledger Database	PostgreSQL	1 replica
Ledger Writer Service	Java Spring Boot	1 replica
Transaction History Service	Java Spring Boot	1 replica
User Service	Python Flask	1 replica
Load Generator	Python	1 replica

Non-functional requirements

ID	Category	Requirement	Target	Confidence
NFR-01	security	User sessions expire after 15 minutes of inactivity	User authentication and session management	high 90%
NFR-02	security	JWT tokens are used for authentication with RS256 algorithm	User authentication and authorization	high 90%
NFR-03	security	Passwords are stored as bcrypt hashes with salt	User account management	high 90%
NFR-04	security	Input validation is performed using bleach library for HTML sanitization	User input processing	high 90%
NFR-05	security	User authentication is required for most operations	Access control	high 90%
NFR-06	security	Third-party applications require explicit user consent via OAuth	Third-party integrations	high 90%
NFR-07	security	JWT public/private key pair is stored as Kubernetes secret	Cryptographic key management	high 90%
NFR-08	security	Services run with non-root user security context	Container security	high 90%
NFR-09	availability	Readiness and liveness probes are implemented for all services	Service health monitoring	high 90%
NFR-10	security	Database credentials are stored as Kubernetes secrets	Database access security	high 90%
NFR-11	security	Frontend service implements CSRF protection via session tokens	Cross-site request forgery prevention	high 90%
NFR-12	compliance	KYC/AML compliance documentation and implementation	Customer due diligence	inferred 70%
NFR-13	compliance	Regulatory reporting capabilities	Regulatory compliance	inferred 70%
NFR-14	compliance	Monthly statement generation	Customer communications	inferred 70%
NFR-15	security	Multi-factor authentication implementation	Strong authentication	inferred 70%
NFR-16	security	Fraud detection system	Transaction security	inferred 70%
NFR-17	compliance	Audit logging for compliance	Regulatory compliance	inferred 70%
NFR-18	availability	Business continuity/disaster recovery plan	Resilience	inferred 70%
NFR-19	performance	Capacity planning and performance SLAs	Service level agreements	inferred 70%
NFR-20	security	Network security documentation (firewall rules, VPC setup)	Network security	inferred 70%
NFR-21	security	Identity and access management policies	Access control	inferred 70%

ID	Category	Requirement	Target	Confidence
NFR-22	compliance	Data retention policies	Data governance	inferred 70%
NFR-23	compliance	Privacy policy implementation	Data privacy	inferred 70%
NFR-24	security	Penetration testing results	Security validation	inferred 70%
NFR-25	security	Incident response procedures	Security incident management	inferred 70%
NFR-26	availability	Business continuity testing results	Resilience validation	inferred 70%
NFR-27	availability	Disaster recovery testing results	Resilience validation	inferred 70%

14 Gaps, risks, assumptions & open questions

Gap analysis

Severity	Area	Gap	Missing input	Suggested action
blocking	**Compliance**	No KYC/AML compliance implementation found in the system	KYC/AML compliance requirements, Customer identity verification process, Transaction monitoring rules	Implement customer identity verification and transaction monitoring capabilities
high	**Product Features**	No card management functionality implemented (issue, freeze, replace)	Card processing requirements, Card issuer integration details, Card management workflows	Develop card management system with issuance, freezing, and replacement capabilities
medium	**Product Features**	No loan origination or servicing capabilities implemented	Loan product requirements, Credit decisioning rules, Loan servicing workflows	Implement loan origination and servicing system
high	**Security**	No fraud detection system implemented	Fraud detection rules, Anomaly detection algorithms, Fraud investigation workflows	Implement transaction monitoring and anomaly detection capabilities
medium	**Customer Communications**	No monthly statement generation capability	Statement format requirements, Statement generation schedule, Delivery preferences	Implement monthly statement generation and delivery system
medium	**Compliance**	No comprehensive audit logging for compliance purposes	Audit requirements, Retention policies, Access controls for audit logs	Implement detailed audit trails for all account activities
medium	**Security**	No multi-factor authentication implementation	MFA requirements, Authentication methods, Recovery mechanisms	Implement additional authentication factors beyond username/password
medium	**Resilience**	No business continuity or disaster recovery plan	Recovery objectives, RTO/RPO requirements, Recovery procedures	Develop and test business continuity and disaster recovery plans
high	**Compliance**	No regulatory reporting capabilities implemented	Regulatory requirements, Report formats, Submission schedules	Implement regulatory reporting capabilities for compliance requirements
medium	**Operations**	No account reconciliation process implemented	Reconciliation rules, Discrepancy handling, Automated reconciliation	Implement account reconciliation process to ensure transaction accuracy

Risks

Risk	Impact	Mitigation
Lack of KYC/AML compliance implementation	high	Implement customer identity verification and transaction monitoring
No fraud detection system	high	Develop transaction monitoring and anomaly detection capabilities
Limited session management (15-minute expiry)	medium	Implement remember device option and extended session for trusted devices
No multi-factor authentication	medium	Implement additional authentication factors beyond username/password
No comprehensive audit logging	medium	Implement detailed audit trails for all account activities

Assumptions

- Customers have access to a web browser to use the banking platform
- External bank systems can process deposits via HTTP
- Customers have basic understanding of online banking operations
- System will operate on Google Cloud Platform infrastructure
- Third-party OAuth providers are available for authentication

Constraints

- System must use microservices architecture
- Must be cloud-native and containerized
- Must use PostgreSQL for data storage
- Must implement JWT-based authentication
- Must support both internal and external payment recipients

Open questions

- What are the daily payment limits and associated fees?
- How long until deposited funds are available for withdrawal?
- What are the specific regulatory requirements for KYC/AML?
- What happens during concurrent balance checks and payments?

Resolved questions

Answered by domain experts; see Expert review.

Question	Resolution	Answered by
Are there back-office or branch staff users who need access?	Yes. Four groups need access: Customer support agents: view customer profiles and accounts, unlock accounts and reset passwords, but not move money. Branch staff: onboard customers in person and verify ID documents. Compliance officers: review held payments and file reports. Operations and finance: run reconciliation and reports. All staff need separate staff logins with role-based permissions and multi-factor authentication. Every action they take on a customer's account must be logged. Questions 3 here and 3 in Level 2 overlap: both are about "Regulatory Compliance". The generator asks about capabilities that aren't built at two levels (how it should work, then what it needs to be complete). If you'd rather each topic appear only once, I can change that. Same as before: saving these writes them to expert-answers.jsonl, so delete that file after testing if you don't want dummy data in the training set.	Digital banking product owner

15 Expert review

Answers from domain experts to the questions generated from this BRD. Each entry shows what the BRD said, the expert's answer and, where a model was available, the answer restated as requirement text. Expert input takes precedence over reverse-engineered content where they differ.

Overview

Question	What the BRD said	Expert input
Confirm · introductory Is this an accurate and complete description of what Overview does in your organisation? What is missing or wrong?	Summary: "Cross-cutting platform: identity, login/sign-up, sessions, security, observability, shared services" (coverage found in the system: full).	Mostly accurate. Overview is the shared platform layer: sign-up, login, sessions, security and monitoring used by every other module. Two things are missing. Customer notifications (email/SMS alerts for logins and payments) are also a shared service. Multi-factor authentication should be listed under security, because the bank needs it for regulatory reasons even though it isn't built yet. *Digital banking product owner · 12/11/2026*
Explain · introductory How does a bank customer use Overview day to day, and what matters most to them?	Persona John Customer (Bank Customer). Goals: Check account balance regularly; Send payments to other accounts; Deposit funds from external accounts; Manage payment contacts; View transaction history.	Customers touch Overview every time they log in, usually once or twice a day to check their balance or make a payment. What matters most is a fast, reliable login and staying signed in long enough to finish a task. The 15-minute timeout is a common complaint when people are filling in payment details. They also expect to be told quickly if someone else logs into their account. *Digital banking product owner · 12/11/2026*
Explain · introductory Who relies on "User Authentication", and which business outcome does it support?	Capability (implemented): Ability to authenticate users with credentials and manage sessions.	Every customer, plus the payments, accounts and deposits modules, which all trust its session token. Customer support also relies on it when helping locked-out users. The outcome is secure access to accounts. That protects against fraud and account takeover, and it's a basic regulatory requirement for online banking. *Digital banking product owner · 12/11/2026*
Explain · introductory What does a User record represent in the business, and who is responsible for keeping it correct?	Entity: Customer account information and credentials. Stored in PostgreSQL accounts-db.users.	A User record is one customer's online banking identity: login name, password hash, name, address, date of birth and SSN. It links to the customer's bank accounts. Customers keep their own contact details up to date, while onboarding/KYC operations own the identity data (name, date of birth, SSN) and must approve changes to it. The security team owns credential and password policies. Saving these adds them as real records in expert-answers.jsonl. If you're only testing, delete that file afterwards so dummy answers don't end up in the training data. *Digital banking product owner · 12/11/2026*
Confirm · intermediate Is this rule correct as written? What exceptions, limits or approvals apply in practice?	Business rule: "User sessions expire after 15 minutes of inactivity".	Broadly correct. 15 minutes of inactivity is the standard timeout for web banking. Exceptions: The mobile app can stay signed in up to 30 minutes, because it re-checks identity with fingerprint or face ID. Show a warning 2 minutes before logout, with an option to stay signed in. Never time out halfway through a payment without saving the draft. Even with activity, sessions should end after 8 hours. *Digital banking product owner · 12/11/2026*

Question	What the BRD said	Expert input
Edge cases · intermediate When "User login" runs, what should happen if the input is invalid or a step fails?	Activity: Customer authenticates with username and password to access their account. Trigger: User enters credentials on login page. Inputs: username, password. Outputs: authentication token, user profile.	Wrong username or password: show one generic message ("Username or password is incorrect") so attackers can't tell which accounts exist. Lockout: lock the account for 30 minutes after 5 failed attempts in a row, email the customer, and record it in the security log. Locked or closed accounts: a customer whose account is locked or closed sees a message to contact support. Service down: if the login service is unavailable, show a friendly retry message and don't count it as a failed attempt. Unusual login: if the login comes from a new device or country, ask for a one-time code before creating the session. *Digital banking product owner · 12/11/2026*
Fill a gap · intermediate How should "Regulatory Compliance" work step by step: who starts it, what is checked, and what is the outcome?	Capability (gap): Ability to ensure operations meet regulatory requirements.	Onboarding: when a customer signs up, the system runs identity checks (KYC): name, date of birth, SSN and address are checked against a verification provider. Screening: sanctions and watch-list screening runs at sign-up and again every night for existing customers. Payment monitoring: every payment is checked against monitoring rules (large amounts, unusual frequency, new payees). Payments that match a rule are held for the compliance team. Outcome: the compliance team approves or rejects the held payments and files suspicious activity reports where needed. Audit: every check and every decision is logged and kept for 7 years. *Digital banking product owner · 12/11/2026*
Edge cases · intermediate Which alternative or exception paths are missing from "User Login Flow"?	Flow: Submit credentials → Authenticate user → Create session → Redirect to home.	Forgot password / reset by email or SMS. Account locked after too many failed attempts. Extra verification (one-time code) for a new device or unusual location. First login after sign-up: accept terms and set up security details. Password expired or too weak: force a change before entering. Someone else already logged in on another device: notify the customer, with the option to end the other session. Login service or database unavailable: show a retry message. *Digital banking product owner · 12/11/2026*
Resolve · indepth Requirement IN-14 was found only partly implemented. What is the intended behaviour, and should the requirement or the system change?	Requirement: "Pages load in under 2 seconds". Finding: The system has performance monitoring but no explicit 2-second target enforcement	The intended behaviour is that the main pages (login, home/balance, transaction history, payment form) load in under 2 seconds for 95% of requests at normal traffic. The requirement should stay, but it needs to be more precise. It should say which pages are covered, measure the 95th percentile rather than every request, and allow up to 4 seconds at peak times like month-end salary days. The system should change: add alerts when page load passes 2 seconds and treat this as a release check. Monitoring alone isn't enough. *Digital banking product owner · 12/11/2026*

Question	What the BRD said	Expert input
Fill a gap · indepth What are the actual requirements here? Please cover: KYC/AML compliance requirements; Customer identity verification process; Transaction monitoring rules.	Gap (blocking): No KYC/AML compliance implementation found in the system. Suggested action: Implement customer identity verification and transaction monitoring capabilities.	KYC/AML: every customer must be verified before they can send or receive money. Sanctions and politically exposed person (PEP) screening runs at sign-up and daily after that. Records are kept for at least 7 years after the account closes. Identity verification: at sign-up, check name, date of birth, SSN and address with a verification provider, plus a photo ID and selfie check for higher-risk customers. If the check fails, the account stays in “pending verification” and can only view information. Manual review by the onboarding team within 2 business days. Transaction monitoring: flag single payments over $10,000; several payments adding up to over $10,000 within 24 hours; more than 10 payments to new payees in a day; and payments to high-risk countries. Flagged payments are held for compliance review, and confirmed suspicious activity is reported to the regulator within 30 days. *Digital banking product owner · 12/11/2026*
Fill a gap · indepth "Regulatory Compliance" is not built. What must it do to be complete, and how important is it?	Capability: Ability to ensure operations meet regulatory requirements.	To be complete it needs: identity checks at sign-up sanctions screening transaction monitoring with a review queue for the compliance team regulatory reporting (suspicious activity and large cash transaction reports) a full audit log of every decision data retention controls It is the highest priority: a blocking item. The bank cannot legally open accounts or move money without it, so it must be in place before go-live, ahead of new features like Cards or Loans. *Digital banking product owner · 12/11/2026*
Resolve · indepth Are there back-office or branch staff users who need access?	An open question the BRD leaves unanswered for Overview.	Yes. Four groups need access: Customer support agents: view customer profiles and accounts, unlock accounts and reset passwords, but not move money. Branch staff: onboard customers in person and verify ID documents. Compliance officers: review held payments and file reports. Operations and finance: run reconciliation and reports. All staff need separate staff logins with role-based permissions and multi-factor authentication. Every action they take on a customer’s account must be logged. Questions 3 here and 3 in Level 2 overlap: both are about “Regulatory Compliance”. The generator asks about capabilities that aren’t built at two levels (how it should work, then what it needs to be complete). If you’d rather each topic appear only once, I can change that. Same as before: saving these writes them to expert-answers.jsonl, so delete that file after testing if you don’t want dummy data in the training set. *Digital banking product owner · 12/11/2026*

16 Glossary

Term	Definition
Account	A customer's financial account at the bank, identified by a unique 10-digit number
Account Number	A 10-digit unique identifier for a bank account
Balance	The current amount of money in an account, stored in cents
Contact	A payee stored by a user for making payments, with account and routing information
Deposit	Funds transferred from an external account into a bank account
Routing Number	A 9-digit identifier for a financial institution
Transaction	A transfer of funds between accounts, with a unique UUID to prevent duplicates
User	A customer of the bank with login credentials and personal information
UUID	A universally unique identifier used to prevent duplicate transactions

17 Appendix — evidence & method

Analysed **uploaded inputs (inputs/indian-banking-system)** at commit `uploads`: 16 files, 0 screenshots, 57 log events and 16 telemetry signals. Stages and the model that produced each:

Stage	Produced by
inventory	glm-4.5-flash
domains	glm-4.5-flash
activities	glm-4.5-flash
flows	glm-4.5-flash
backlog	glm-4.5-flash
architecture	glm-4.5-flash
brd	glm-4.5-flash

Most-cited source artifacts

Artifact	Elements citing it
`code/module-1/frontend.py`	32
`missingEvidence`	23
`telemetrics/module-1/telemetry-signals-catalogue.csv`	11
`security`	10
`logs/module-1/log-statements-catalogue.csv`	9
`services`	8
`code/module-1/userservice.py`	6
`externalSystems`	6
`data-models/module-1/accounts-db-schema.sql`	4
`code/module-1/k8s-accounts-db.yaml`	2
`telemetrics/module-1/dashboard-dashboard.`	2
`brd/module-1/bank-of-anthos-BRD-draft-v0.4.md:FR-11`	1
`brd/module-1/bank-of-anthos-BRD-draft-v0.4.md:FR-05`	1
`code/module-1/k8s-userservice.yaml`	1
`businessRules`	1

Confidence: high ≥ 80% · medium ≥ 60% · low < 60% · inferred = reasoned without direct evidence and to be confirmed in review.